

This document contains a full dump of all questions in the selected question banks. Questions begin on the next page.

1. (1) What is NaCl commonly known as?
 - a. Water
 - b. Salt**
 - c. Sugar
 - d. Oxygen
2. **T** True / False (F) (2) The speed of light is approximately 3.0×10^8 meters per second.
3. (3) Which organelle is known as the powerhouse of the cell?
 - a. Nucleus
 - b. Mitochondria**
 - c. Ribosome
 - d. Endoplasmic Reticulum
4. **T** True / False (F) (4) Earth is the third planet from the Sun in our solar system.
5. (5) What is the chemical symbol for gold?
 - a. Au**
 - b. Ag
 - c. Fe
 - d. Pb
6. (8) What is the atomic number of Carbon?
 - a. 4
 - b. 6**
 - c. 8
 - d. 12
7. (9) Which element has the chemical symbol 'O'?
 - a. Osmium
 - b. Oxygen**
 - c. Oganesson
 - d. Oxide
8. (10) What type of bond involves the sharing of electron pairs?
 - a. Ionic bond
 - b. Covalent bond**
 - c. Metallic bond
 - d. Hydrogen bond

9. (11) What is the molecular formula for water?

- a. H₂O c. O₂
- b. CO₂ d. H₂O₂

10. (12) Which gas is most abundant in Earth's atmosphere?

- a. Oxygen **c**. Nitrogen
b. Carbon Dioxide d. Hydrogen

11. (13) What is the name of the process where a solid turns directly into a gas?

- a. Evaporation
b. Sublimation
c. Condensation
d. Deposition

12. (14) Which element is the most electronegative?

- a. Oxygen
b. Fluorine
c. Nitrogen
d. Chlorine

13. (15) What is the molar mass of carbon dioxide (CO_2)?

- a. 28 g/mol **c**. 44 g/mol
b. 32 g/mol d. 64 g/mol

14. (16) Which of the following is a noble gas?

- a. Oxygen c. Helium
- b. Nitrogen d. Hydrogen

15. (17) What is the charge of a proton?

- a. Negative c. Neutral
b. Positive d. Variable

16. True (T) / **F** False (18) The periodic table is arranged by increasing atomic mass.

17. ☒ True / ☐ False (F) (19) Hydrogen is the lightest element.
18. True (T) / ☒ False (20) Covalent bonds are formed by the transfer of electrons.
19. ☒ True / ☐ False (F) (21) Catalysts speed up chemical reactions without being consumed.
20. True (T) / ☒ False (22) The noble gases are highly reactive.
21. ☒ True / ☐ False (F) (23) Water boils at 100 degrees Celsius at standard atmospheric pressure.
22. True (T) / ☒ False (24) Isotopes of an element have different numbers of protons.
23. ☒ True / ☐ False (F) (25) A base has a pH level greater than 7.
24. True (T) / ☒ False (26) Electrons have a positive charge.
25. ☒ True / ☐ False (F) (27) Carbon can form four covalent bonds.
26. (6) What is the pH level of pure water?
 - a. 5
 - b. 7
 - c. 9
 - d. 11
27. ☒ True / ☐ False (F) (7) An acid has a pH level less than 7.
28. (28) What is the SI unit of force?
 - a. Joule
 - b. Newton
 - c. Watt
 - d. Pascal

29. (29) What is the acceleration due to gravity on Earth?
- a. 8.9 m/s^2
 - b. 9.8 m/s^2**
 - c. 10.8 m/s^2
 - d. 11.8 m/s^2
30. (30) Which law states that for every action there is an equal and opposite reaction?
- a. Newton's First Law
 - c. Newton's Third Law**
 - b. Newton's Second Law
 - d. Law of Conservation of Energy
31. (31) What is the SI unit of electric current?
- a. Volt
 - b. Ampere**
 - c. Ohm
 - d. Coulomb
32. (32) What type of energy is stored in a stretched spring?
- a. Kinetic energy
 - b. Potential energy**
 - c. Thermal energy
 - d. Chemical energy
33. (33) What is the formula for calculating kinetic energy?
- a. $KE = mv$
 - b. $KE = \frac{1}{2} mv^2$**
 - c. $KE = mgh$
 - d. $KE = Fd$
34. (34) Which particle in an atom has no electric charge?
- a. Proton
 - c. Neutron**
 - b. Electron
 - d. Photon
35. (35) What is the unit of electrical resistance?
- a. Ampere
 - c. Ohm**
 - b. Volt
 - d. Watt

36. (36) What phenomenon causes light to bend when passing from one medium to another?
- a. Reflection
 - ☒ b. Refraction
 - c. Diffraction
 - d. Interference
37. (37) What is the frequency of a wave with a period of 0.5 seconds?
- a. 0.5 Hz
 - ☒ c. 2 Hz
 - b. 1 Hz
 - d. 4 Hz
38. True (T) / ☒ False (38) Energy can be created or destroyed according to the law of conservation of energy.
39. ☒ True / False (F) (39) Sound waves require a medium to travel through.
40. True (T) / ☒ False (40) The speed of sound is faster than the speed of light.
41. ☒ True / False (F) (41) Momentum is conserved in a closed system.
42. True (T) / ☒ False (42) A concave lens converges light rays.
43. ☒ True / False (F) (43) Temperature is a measure of the average kinetic energy of particles.
44. ☒ True / False (F) (44) An object at rest will remain at rest unless acted upon by an external force.
45. ☒ True / False (F) (45) Magnetic field lines always form closed loops.
46. True (T) / ☒ False (46) The wavelength of visible light is longer than that of radio waves.
47. ☒ True / False (F) (47) Power is the rate at which work is done.

48. (48) What is the basic unit of life?

- a. Atom
- b. Molecule
- ☒ c. Cell
- d. Organ

49. (49) What is the process by which plants make their own food?

- a. Respiration
- b. ☒ Photosynthesis
- c. Fermentation
- d. Digestion

50. (50) Which molecule carries genetic information?

- a. RNA
- b. Protein
- ☒ c. DNA
- d. Lipid

51. (51) What type of organism is a mushroom?

- a. Plant
- b. Animal
- ☒ c. Fungus
- d. Bacteria

52. (52) Which blood cells are responsible for carrying oxygen?

- a. White blood cells
- b. ☒ Red blood cells
- c. Platelets
- d. Plasma cells

53. (53) What is the largest organ in the human body?

- a. Heart
- b. Liver
- ☒ c. Skin
- d. Brain

54. (54) Which system is responsible for breaking down food?

- a. Respiratory system
- b. Circulatory system
- ☒ c. Digestive system
- d. Nervous system

55. (55) What is the term for an organism that can make its own food?
- a. Heterotroph
 - b. Autotroph**
 - c. Decomposer
 - d. Consumer
56. (56) Which part of the cell contains the genetic material?
- a. Cytoplasm
 - b. Nucleus**
 - c. Cell membrane
 - d. Vacuole
57. (57) What is the function of chlorophyll in plants?
- a. Water absorption
 - b. Light absorption**
 - c. Oxygen production
 - d. Nutrient storage
58. **T** True / False (F) (58) Humans have 46 chromosomes in each body cell.
59. True (T) / **F** False (59) Bacteria are eukaryotic organisms.
60. **T** True / False (F) (60) The heart pumps blood throughout the body.
61. **T** True / False (F) (61) All living organisms are made up of cells.
62. True (T) / **F** False (62) Mitosis results in four daughter cells.
63. **T** True / False (F) (63) Antibodies are produced by white blood cells.
64. True (T) / **F** False (64) Photosynthesis occurs in the mitochondria.
65. **T** True / False (F) (65) The brain is part of the central nervous system.
66. **T** True / False (F) (66) Enzymes are proteins that catalyze chemical reactions.

67. True (T) / ☒ False (67) Viruses are considered living organisms.

68. (68) What is the largest planet in our solar system?

- a. Saturn
- ☒ b. Jupiter
- c. Neptune
- d. Uranus

69. (69) What is the name of Earth's natural satellite?

- a. Phobos
- ☒ c. Moon
- b. Titan
- d. Europa

70. (70) Which planet is known as the Red Planet?

- a. Venus
- ☒ b. Mars
- c. Mercury
- d. Jupiter

71. (71) What is the closest star to Earth?

- a. Alpha Centauri
- ☒ c. The Sun
- b. Sirius
- d. Proxima Centauri

72. (72) What type of galaxy is the Milky Way?

- a. Elliptical
- ☒ b. Spiral
- c. Irregular
- d. Lenticular

73. (73) What is a light-year a measure of?

- a. Time
- ☒ b. Distance
- c. Speed
- d. Brightness

74. (74) Which planet has the most moons?

- a. Jupiter
- ☒ b. Saturn
- c. Uranus
- d. Neptune

75. (75) What is the name of the force that keeps planets in orbit around the Sun?

- a. Magnetism
- ☒ b. Gravity
- c. Nuclear force
- d. Friction

76. (76) What is a supernova?

- a. A new star
- ☒ b. An exploding star
- c. A black hole
- d. A nebula

77. (77) Which planet is tilted on its side?

- a. Saturn
- ☒ c. Uranus
- b. Neptune
- d. Venus

78. ☒ T True / False (F) (78) The Sun is a star.

79. True (T) / ☒ F False (79) Mercury is the hottest planet in our solar system.

80. True (T) / ☒ F False (80) A lunar eclipse occurs when the Moon passes between the Earth and the Sun.

81. ☒ T True / False (F) (81) Saturn is known for its prominent ring system.

82. ☒ T True / False (F) (82) Black holes have such strong gravity that not even light can escape.

83. ☒ T True / False (F) (83) The asteroid belt is located between Mars and Jupiter.

84. True (T) / ☒ F False (84) Pluto is still classified as a major planet.

85. ☒ T True / False (F) (85) Comets are made of ice, dust, and rocky material.

86. ☒ T True / False (F) (86) The universe is approximately 13.8 billion years old.

87. True (T) / ☒ False (87) Neptune is closer to the Sun than Uranus.
88. (88) What is the largest organ in the human body?
- a. Liver
 - ☒ c. Skin
 - b. Brain
 - d. Heart
89. (89) What is the normal human body temperature in Celsius?
- a. 35°C
 - ☒ b. 37°C
 - c. 39°C
 - d. 40°C
90. (90) Which vitamin is produced when skin is exposed to sunlight?
- a. Vitamin A
 - ☒ c. Vitamin D
 - b. Vitamin C
 - d. Vitamin K
91. (91) What is the medical term for high blood pressure?
- a. Hypotension
 - ☒ b. Hypertension
 - c. Hyperglycemia
 - d. Hypoglycemia
92. (92) Which organ produces insulin?
- a. Liver
 - ☒ c. Pancreas
 - b. Kidney
 - d. Stomach
93. (93) What is the main function of red blood cells?
- a. Fighting infection
 - ☒ c. Carrying oxygen
 - b. Clotting blood
 - d. Producing hormones
94. (94) Which part of the brain controls balance and coordination?
- a. Cerebrum
 - ☒ b. Cerebellum
 - c. Medulla
 - d. Hypothalamus

95. (95) What is the medical term for a heart attack?

- a. Cardiac arrest
- ☒ b. Myocardial infarction
- c. Angina pectoris
- d. Arrhythmia

96. (96) Which blood type is considered the universal donor?

- a. A
- b. B
- c. AB
- ☒ d. O

97. (97) What is the main function of the kidneys?

- a. Digesting food
- ☒ b. Filtering blood
- c. Pumping blood
- d. Producing bile

98. True (T) / ☒ False (F) (98) Antibiotics are effective against viral infections.

99. ☒ True (T) / False (F) (99) The human body has 206 bones in adulthood.

100. ☒ True (T) / False (F) (100) Vaccination helps the body develop immunity to diseases.

101. ☒ True (T) / False (F) (101) The liver can regenerate itself.

102. True (T) / ☒ False (F) (102) Type 1 diabetes is caused by lifestyle factors such as poor diet.

103. True (T) / ☒ False (F) (103) The appendix has no known function in the human body.

104. ☒ True (T) / False (F) (104) Aspirin is commonly used to reduce pain and inflammation.

105. True (T) / ☒ False (F) (105) The spinal cord is part of the peripheral nervous system.

106. ☒ True (T) / False (F) (106) Dehydration can cause fatigue and dizziness.

107. True (T) / ☒ False (107) Cholesterol is always harmful to the body.

108. (108) What are the three main layers of the Earth?

- ☒ a. Crust, mantle, core
- b. Lithosphere, asthenosphere, mesosphere
- c. Inner core, outer core, crust
- d. Soil, rock, magma

109. (109) What is the process by which water changes from a liquid to a gas?

- a. Condensation
- ☒ b. Evaporation
- c. Precipitation
- d. Sublimation

110. (110) Which type of plate boundary occurs when two plates move apart?

- a. Convergent
- ☒ b. Divergent
- c. Transform
- d. Subduction

111. (111) What is the most abundant gas in Earth's atmosphere?

- a. Oxygen
- ☒ b. Nitrogen
- c. Carbon dioxide
- d. Argon

112. (112) Which scale is used to measure earthquake intensity?

- ☒ a. Richter scale
- b. Beaufort scale
- c. Mohs scale
- d. Fujita scale

113. ☒ True / False (F) (113) Weathering is the breakdown of rocks by physical or chemical processes.

114. True (T) / ☒ False (114) Sedimentary rocks are formed from molten magma.

115. True (T) / ☒ False (115) The mantle is the hottest layer of the Earth.

116. ☒ True / False (F) (116) Fossils are used to determine the age of rocks and the history of life on Earth.

117. ☒ True / False (F) (117) Ocean trenches are formed at convergent plate boundaries.

End of Exam